

DESA ADITHYA

Software Engineer — AI Researcher — Cloud Enthusiast

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SUMMARY

Final-year CS undergraduate (CGPA 9.17/10) with published IEEE research and hands-on experience in federated learning, IoT systems, full-stack development, and cloud platforms (AWS, Azure). Proven ability to deliver quantified, deployable solutions across ML and software engineering domains.

EDUCATION

Vardhaman College of Engineering

B.Tech in Computer Science & Engineering **CGPA: 9.17 / 10.0**

Hyderabad, Telangana

Aug 2023 – Apr 2027

– Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning, Cloud Computing

Pragathi Junior College, Telangana State Board of Intermediate Education (TSBIE)

Class XII (MPC) **98.7%**

Telangana

2023

SPR High School, Board of Secondary Education, Telangana

Class X **CGPA: 10.0 / 10.0**

Telangana

2021

TECHNICAL SKILLS

Core CS Concepts: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Machine Learning, Deep Learning

Web Technologies: HTML, CSS, JavaScript, JSP, Servlets, REST APIs

Databases: MySQL, SQL

Cloud Platforms: AWS, Microsoft Azure

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Federated Learning, Neural Networks, Graph CNN

Developer Tools: Git, Linux, Android Studio, Arduino IDE

Software Engineering: OOP, Agile, SDLC, CI/CD

Hardware & IoT: Arduino, GPS, GSM, Embedded Systems

PROJECTS

AquaVigil – IoT Flood Monitoring & Emergency Response System Arduino | C | IoT | GPS | GSM *Feb 2025 – Apr 2025*

- Designed and developed an IoT-based flood monitoring and emergency response system using Arduino, GPS, and GSM modules.
- Implemented a three-level flood alert mechanism triggering alarms, automated SMS notifications, and emergency calls.
- Enabled real-time location sharing with rescue services during critical flood conditions to support disaster-response operations.
- Engineered reliable power management to ensure uninterrupted operation during extreme weather events.

Energy-Efficient Mobile Sink Path Planning in HWSNs Python | Federated Learning | Graph CNN *Dec 2025 – Feb 2026*

- Developed a Federated Graph CNN framework for mobile sink path optimization in Heterogeneous Wireless Sensor Networks (HWSNs).
- Modeled wireless sensor nodes as graph structures and implemented privacy-preserving federated learning across distributed clients.
- Improved network lifetime from 1200 to 1650 rounds while increasing load-balancing efficiency from 72% to 89%.
- Enhanced packet delivery ratio from 88% to 95% through adaptive sink-routing optimization.

Federated Learning – Heart Disease Detection with Concept Drift PyTorch | Federated Learning *May 2025 – Aug 2025*

- Developed a privacy-preserving federated learning framework for heart disease prediction by sharing model gradients instead of sensitive patient records.

- Implemented adaptive concept-drift detection mechanisms to automatically retrain models under changing data distributions, improving long-term predictive reliability.

DeepFake Image Detection System CNN | Transfer Learning | PyTorch

Mar 2026 – May 2026

- Built a deep learning-based image classification system to identify AI-generated and manipulated media using transfer learning techniques.
- Compared multiple CNN architectures and optimized inference performance for robust deepfake detection across diverse image datasets.

CERTIFICATIONS

Microsoft Azure SQL · Microsoft · Coursera

Nov 2025

Responsible AI: Applying AI Principles with Google Cloud · Google Cloud · Coursera

Nov 2025

Introduction to Artificial Intelligence (AI) · IBM · Coursera

Nov 2025

Data Analytics Job Simulation · Deloitte · Forage

Jul 2025

Linguaskill Business – B2 (Listening: C1) · Cambridge University Press & Assessment

Mar 2026

PUBLICATIONS & RESEARCH

EchoSync: Smart Audio Recommendation Engine Python | Neural Networks | IEEE INDISCON 2025

Aug 2025

- Co-authored and presented research on neural vocal mapping and mood-aware music recommendation systems at IEEE INDISCON 2025 hosted at NIT Rourkela. DOI: 10.1109/INDISCON66021.2025.11251708

Real-Time MTCNN-Based Face Recognition for Criminal & Missing Person ID Python | MTCNN | ICAICCIT Nov 2025

- Presented a real-time MTCNN-based face recognition system for criminal and missing-person identification at ICAICCIT 2025. DOI: 10.1109/ICAICCIT68829.2025.11434096

Automatic Diagnosis of Schizophrenia Using Hybrid Neural Networks Python | scikit-learn | IJIRT Aug 2025

- Published research on automated schizophrenia diagnosis using hybrid neural network architectures in IJIRT (Vol. 12, Issue 3). ijirt.org/article?manuscript=183624

ACHIEVEMENTS & LEADERSHIP

Academic Excellence

Maintained a CGPA of 9.17/10, ranked among the top-performing students in the Computer Science cohort at Vardhaman College of Engineering.

Secretary, Abhinaya Club (Drama Club)

Jul 2024 – Jan 2025

Managed club communications, maintained production records, and coordinated scheduling for a 10+ member team executing drama productions and cultural events.

Volunteer Lead, Ortus 2K24 Annual Fest

2024

Managed end-to-end execution of the Gully Cricket event with 100+ participants, overseeing logistics and on-ground coordination.